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Medan Elektromagnetik

Latihan

1) Sebuah material dielektrik dikenai medan listrik dengan persamaan $\vec{E} = 0,1 \cos(2\pi \times 10^6 t)\, \vec{a}_v$ V/m, material tersebut adalah polystyrene dengan $\varepsilon_r = 2,5$ dan distilled water $\varepsilon_r = 81$. Tentukan

a) Vektor polarisasi
b) Suseptibilitas elektrik
c) Rapat arus polarisasi
d) Rapat muatan polarisasi

* Polystyrene

a) $\bar{P} = \varepsilon_0 \chi_e \vec{E}$

$= \varepsilon_0 . (\varepsilon_r - 1) . \vec{E}$

$= \frac{1}{36\pi} \times 10^{-9} \times (2,5 - 1) \times 0,1 \cos(2\pi \times 10^6 t)\, \bar{a}_v$

$= 1,3 \times 10^{-12} \cos(2\pi \times 10^6 t)\, \bar{a}_v$

b) $\chi_e = \varepsilon_r - 1$

$= 2,5 - 1$

$= 1,5$

c) $\bar{J}_p = \frac{\partial \bar{P}}{\partial t}$

$= \frac{\partial (1,3 . 10^{-12} \cos(2\pi \times 10^6 t)\, \bar{a}_v)}{\partial t}$

$= -2,6\pi . 10^{-6} \sin(2\pi . 10^6 t)\, a_v$

$= -8,168 . 10^{-6} \sin(2\pi . 10^6 t)\, a_v$

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d) $\rho_p = -\nabla \cdot \bar{P}$ $\quad \bar{P} = 1{,}3 \cdot 10^{-12} \cos(2\pi \cdot 10^6 t)\ \bar{a}_x$

$= -\left(\frac{\partial P_x}{\partial x} + \frac{\partial P_y}{\partial y} + \frac{\partial P_z}{\partial z}\right)$ $\quad P_x = 1{,}3 \cdot 10^{-12} \cos(2\pi \cdot 10^6 t)$

$P_y = 0$

$= -\left(\frac{\partial (1{,}3 \cdot 10^{-12} \cos(2\pi \cdot 10^6 t)}{\partial x} + \frac{0}{\partial y} + \frac{0}{\partial z}\right)$ $\quad P_z = 0$

$= 0$

* dielektrik distele water

a) $\bar{P} = \varepsilon_0 \chi_e \bar{E}$

$= \frac{1}{36\pi} \times 10^{-9} \cdot (81-1) \cdot 0{,}1 \cos(2\pi \cdot 10^6 t)\ \bar{a}_x$

$= 7{,}07 \cdot 10^{-11} \cos(2\pi \cdot 10^6 t)\ \bar{a}_x$

b) $\chi_e = \varepsilon_r - 1$

$= 80 - 1$

$= 80$

c) $\bar{J}_p = \frac{\partial P}{\partial t}$

$= \frac{\partial (7{,}07 \cdot 10^{-11} \cos(2\pi \cdot 10^6 t)\ \bar{a}_x)}{\partial t}$

$= -14{,}14\pi \cdot 10^{-5} \sin(2\pi \cdot 10^6 t)\ \bar{a}_x$

$= -4{,}44 \cdot 10^{-4} \sin(2\pi \cdot 10^6 t)\ \bar{a}_x$

d) $\rho_p = -\nabla \cdot \bar{P}$ $\quad \bar{P} = 7{,}07 \cdot 10^{-11} \cos(2\pi \cdot 10^6 t)\ \bar{a}_x$

$= -\left(\frac{\partial P_x}{\partial x} + \frac{\partial P_y}{\partial y} + \frac{\partial P_z}{\partial z}\right)$ $\quad P_x = 7{,}07 \cdot 10^{-11} \cos(2\pi \cdot 10^6 t)$

$P_y = 0$

$= -\left(\frac{\partial (7{,}07 \cdot 10^{-11} \cos(2\pi \cdot 10^6 t)}{\partial x} + \frac{0}{\partial z} + \frac{0}{\partial z}\right)$ $\quad P_z = 0$

$= 0$

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2) Dua buah konduktor masing-masing memiliki konduktivitas berturut turut $\sigma_1 = 5,7 \times 10^7$ S/M dan $\sigma_2 = 3,2 \times 10^7$ S/M Jika kedua konduktor ini dikenai medan listrik $\vec{E} = 10\,\vec{a}_y$ (V/m), tentukan perbandingan rapat arus konduksi konduktor 1 terhadap 2

$$\bar{J}_{c_1} : \bar{J}_{c_2}$$
$$\sigma_1 \bar{E} : \sigma_2 \bar{E}$$
$$\sigma_1 : \sigma_2$$
$$5,7.10^7 : 3,2.10^7$$
$$5,7 : 3,2$$
$$57 : 32$$

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